

tcpdump + Wireshark: Combined Quick Reference

Your One-Page Cheat Sheet for Kerberos Traffic Analysis

Print this. Laminate it. Keep it by your keyboard.

Complete Workflow

```
1. CAPTURE (tcpdump) → 2. ANALYZE (Wireshark) → 3. REPORT
```

CAPTURE (tcpdump)

Basic Capture Template

```
sudo tcpdump -i any -nn -tttt port 88 -w capture_name.pcap
```

Quick Captures by Scenario

Brute Force:

```
sudo tcpdump -i any port 88 -w bruteforce_$(date +%Y%m%d_%H%M%S).pcap
```

AS-REP Roasting:

```
sudo tcpdump -i any port 88 -w asrep_$(date +%Y%m%d_%H%M%S).pcap
```

Kerberoasting:

```
sudo tcpdump -i any port 88 -w kerberoast_$(date +%Y%m%d_%H%M%S).pcap
```

Specific Attacker:

```
sudo tcpdump -i any port 88 and host 192.168.88.10 -w attacker.pcap
```

Stop Capture

```
Ctrl+C
```

Verify Capture

```
tcpdump -r capture.pcap | wc -l      # Count packets
tcpdump -r capture.pcap -c 5         # Preview first 5
```

 ANALYZE (Wireshark)

Open Capture

```
wireshark capture.pcap &
```

Essential Filters

Purpose	Filter
All Kerberos	kerberos
Failed Auth	kerberos.error_code == 24
Successful Auth	kerberos.msg_type == 11 && !kerberos.error_code

Purpose	Filter
AS-REQ	<code>kerberos.msg_type == 10</code>
TGS-REQ	<code>kerberos.msg_type == 12</code>
From Attacker	<code>kerberos && ip.src == 192.168.88.10</code>

Quick Analysis Steps

1. Find Attacker IP:

Statistics > Conversations > TCP tab > Sort by Packets

2. Count Attack Packets:

Apply filter → Note packet count at bottom

3. Follow Attack:

Right-click packet → Follow > TCP Stream

4. Extract Evidence:

File > Export Packet Dissections > As Plain Text



ATTACK SIGNATURES

Brute Force

tcpdump capture:

```
sudo tcpdump -i any port 88 -w bruteforce.pcap
# Run attack script
# Ctrl+C
```

Wireshark analysis:

1. Filter: `kerberos.error_code == 24`
2. Count packets (failed attempts)
3. Statistics > Conversations (find attacker IP)
4. Check success: `kerberos.msg_type == 11 && !kerberos.error_code`

Signature:

- Many error_code 24 from single IP
 - Same username (cname)
 - Regular timing (1-5 per second)
-

AS-REP Roasting

tcpdump capture:

```
sudo tcpdump -i any port 88 -w asrep.pcap
# Run AS-REP roast
# Ctrl+C
```

Wireshark analysis:

1. Filter: `kerberos.msg_type == 11`
2. Look for: `!kerberos.pa_data` (no pre-auth)
3. Count unique usernames tested
4. Find roastable accounts

Signature:

- Multiple AS-REP without pre-auth data
 - Different usernames from same IP
 - Short time window
-

Kerberoasting

tcpdump capture:

```
sudo tcpdump -i any port 88 -w kerberoast.pcap
# Run Kerberoasting
# Ctrl+C
```

Wireshark analysis:

```
1. Filter: kerberos.msg_type == 12
2. Count TGS-REQ packets
3. List unique service names (sname)
4. Check timing pattern
```

Signature:

- Many TGS-REQ from single IP
- Different service principals
- Rapid succession (10+ requests in <1 minute)



One-Liner Commands

Capture + Count

```
sudo tcpdump -i any port 88 -w test.pcap -c 100 # Capture exactly 100 packets
```

Capture + Display

```
sudo tcpdump -i any port 88 -w test.pcap -l | tee test.log # See + save
```

Capture with Auto-Rotate

```
sudo tcpdump -i any port 88 -w capture.pcap -C 100 -W 5 # 100MB files, keep 5
```

Quick Analysis

```
tcpdump -r capture.pcap -nn | grep "error" # Find errors quickly
```

Attack Analysis Checklist

For each capture, document:

[] Basic Info

- Date/time:
- Capture file:
- Total packets:

[] Attack Details

- Type: (brute force / AS-REP / Kerberoast / other)
- Source IP:
- Target: (user/service)
- Duration:

[] Metrics

- Total attempts:
- Failed:
- Successful:
- Rate (per second):

[] Evidence

- Screenshot location:
- Exported text location:
- Key findings:

Common Wireshark Menu Actions

Action	Menu Path
Open File	File > Open

Action	Menu Path
Export Text	File > Export Packet Dissections > As Plain Text
Find Packet	Edit > Find Packet
Follow Stream	Right-click > Follow > TCP Stream
Conversations	Statistics > Conversations
Protocol Stats	Statistics > Protocol Hierarchy
Time Format	View > Time Display Format > Time of Day
Expert Info	Analyze > Expert Information

Portfolio Screenshot Checklist

For each attack type, capture:

[] Overview Screenshot

- Full Wireshark window
- Filter applied
- Packet count visible
- Annotate: Attack type, count, duration

[] Detail Screenshot

- Expanded packet showing attack details
- Highlight key fields (error code, username, etc.)
- Annotate: What makes this malicious

[] Statistics Screenshot

- Conversations or Protocol Hierarchy
- Show attacker IP prominence
- Annotate: Traffic volume comparison

[] Timeline Screenshot

- Time column visible

- Show pattern over time
- Annotate: Attack rate/timing

File Organization Template

```
kerberos-captures/  
├── 20250207/  
│   ├── 01_bruteforce_admin_093042.pcap  
│   ├── 01_bruteforce_admin_093042_analysis.txt  
│   ├── 01_bruteforce_admin_093042.md  
│   ├── 02_asrep_roast_104523.pcap  
│   ├── 02_asrep_roast_104523_analysis.txt  
│   ├── 02_asrep_roast_104523.md  
│   └── screenshots/  
│       ├── bruteforce_overview.png  
│       ├── bruteforce_detail.png  
│       └── bruteforce_stats.png  
└── scripts/  
    ├── capture_attack.sh  
    └── analyze_attack.sh
```

Troubleshooting Quick Fixes

Problem	Solution
No packets captured	Check <code>sudo netstat -tulpn grep :88</code>
Permission denied	Use <code>sudo tcpdump</code>
Can't see Kerberos	Apply filter: <code>kerberos</code>
Too many packets	Filter by IP: <code>kerberos && ip.src == X.X.X.X</code>
Time confusing	View > Time Display Format > Time of Day

Problem	Solution
Lost in data	Start: Statistics > Conversations

Speed Tips

Capture faster:

```
alias kcap='sudo tcpdump -i any port 88 -w'  
# Usage: kcap attack.pcap
```

Analyze faster:

```
alias kshark='wireshark'  
# Usage: kshark attack.pcap &
```

Count faster:

```
alias kcount='tcpdump -r'  
# Usage: kcount attack.pcap | wc -l
```

Add to ~/.bashrc

Mastery Test

You're competent when you can:

1. **Capture** Kerberos traffic to a pcap file
2. **Verify** packet count without opening Wireshark
3. **Open** capture in Wireshark
4. **Filter** to show only failed authentications
5. **Find** the attacker's IP address
6. **Count** total attack attempts
7. **Calculate** attack rate (attempts/second)

8. **Extract** evidence to text file
9. **Take** portfolio-quality screenshot
10. **Document** findings in markdown

Time to competence: 2-3 hours practice

Quick Help

tcpdump man page:

```
man tcpdump
```

Wireshark help:

```
Help > Contents
```

Display filter reference:

```
Help > Supported Protocols
```

Online:

- tcpdump: <https://www.tcpdump.org/>
 - Wireshark: <https://www.wireshark.org/docs/>
-

Your First Hour Plan

Minute 0-15: Setup

- Install tools
- Create directories
- Save this cheat sheet

Minute 15-30: Capture Practice

- Run 3 test captures
- Verify each one
- Count packets

Minute 30-45: Analysis Practice

- Open each capture
- Apply 5 essential filters
- Take 1 screenshot each

Minute 45-60: Real Attack

- Capture a brute force
 - Analyze it completely
 - Document findings
-

Daily Workflow

Morning: Capture

```
cd ~/kerberos-captures/$(date +%Y%m%d)
sudo tcpdump -i any port 88 -w attack_$(date +%H%M%S).pcap
# Run attack script
# Ctrl+C
```

Afternoon: Analyze

```
wireshark attack_*.pcap
# Apply filters
# Take screenshots
# Export evidence
```

Evening: Document

```
# Analysis: [attack_name]
- Type: [brute force/AS-REP/Kerberoast]
- Source: [IP]
- Findings: [summary]
- Evidence: [screenshots]
```



Success Metrics

After 1 week:

- ☐ 10+ captures created
- ☐ All attack types analyzed
- ☐ Screenshots for portfolio
- ☐ Documentation complete

After 2 weeks:

- ☐ Can capture/analyze in <10 minutes
 - ☐ Created custom filters
 - ☐ Automated capture scripts
 - ☐ Portfolio-ready
-



Save This

```
# Save to your home directory
cp tcpdump_wireshark_quickref.md ~/
cat ~/tcpdump_wireshark_quickref.md
```

Or:

1. Print this page
 2. Laminate it
 3. Keep by keyboard
 4. Reference constantly
-

You now have everything needed for professional packet analysis.

Next: Capture your first attack and analyze it!

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 Remember:

- **Capture:** `sudo tcpdump -i any port 88 -w file.pcap`
- **Analyze:** `wireshark file.pcap`
- **Filter:** `kerberos` + specific filters
- **Document:** Screenshots + notes

That's it. You're ready.